ERA 5 calibration to Elexon power

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1 Low wind events

Bias

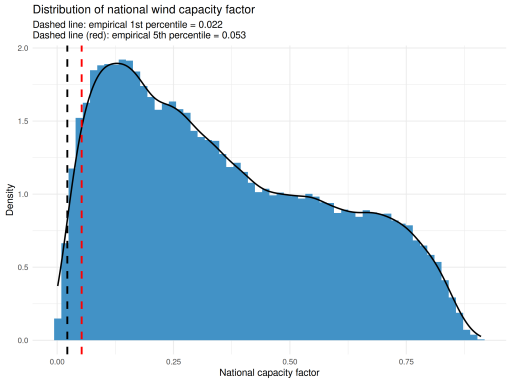
Low wind events

Low wind events

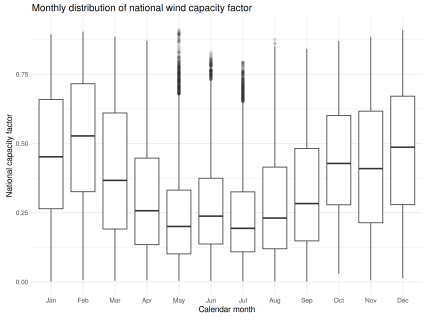
- Important at aggregate GB level for system operation and security of supply.
- We define low wind events as periods of time when the total GB wind power generation is below a certain threshold (e.g., 5% of installed capacity).
- We can analyze the distribution of these events in terms of their duration, frequency, and severity (e.g., cumulative shortfall).
- We can also compare the observed low wind events with those simulated by the ERA5 reanalysis data to assess the accuracy of the reanalysis in capturing these events.
- Ideally we'll model shortfall events, considering demand and other generation sources, but for now we focus on wind generation only.

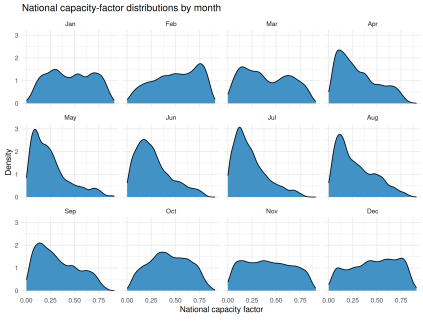
Capacity Factor distribution

- Historical distribution of national capacity factor (CF) for wind power in GB.
- The CF is defined as the ratio of actual power output to the maximum possible output in one hour
- Data spans from 2019 up to 2025 for 147 locations across GB.

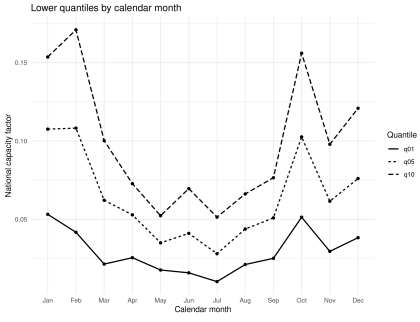


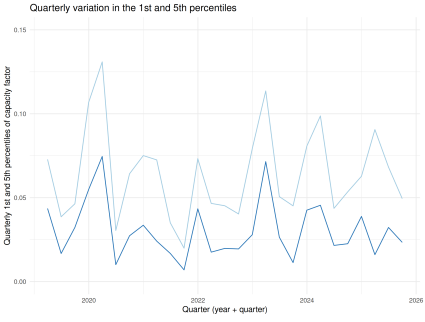
Seasonal variation of CF



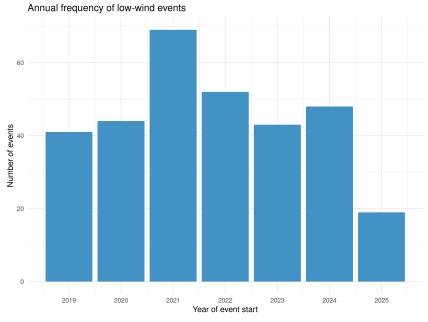


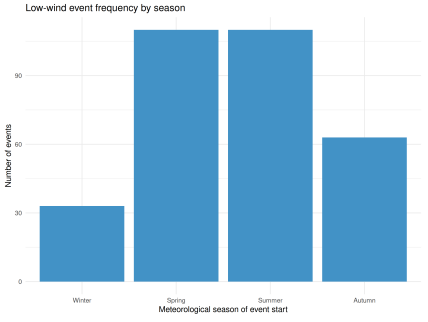
CF quantiles seasonality



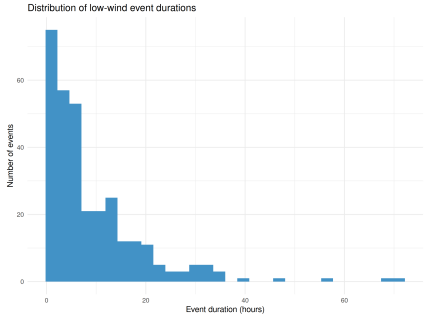


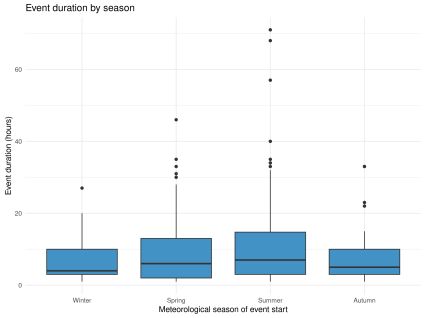
Frequency of events below 5% CF



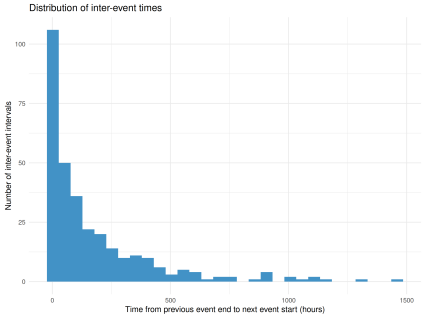


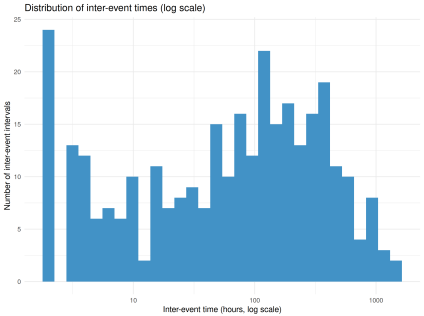
Duration of events below 5% CF



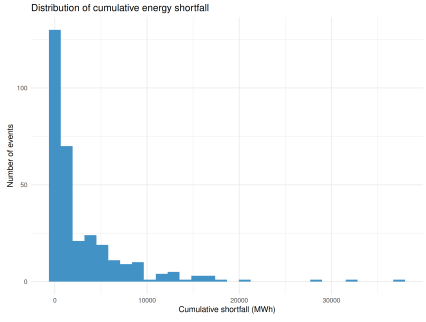


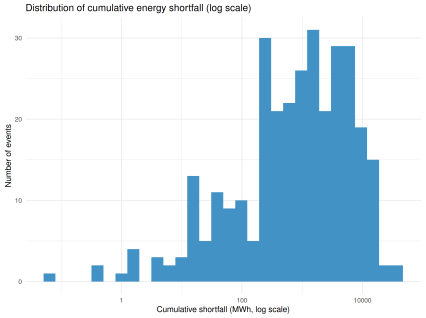
Interevent time of events below 5% CF



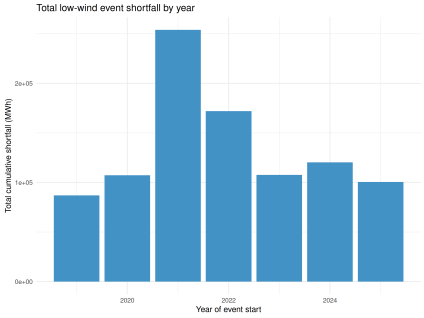


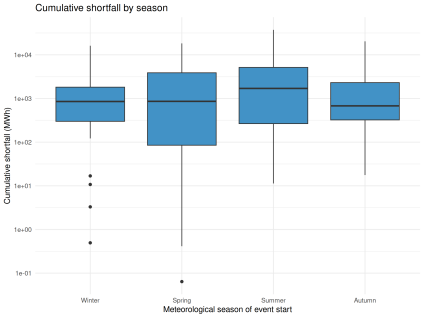
Severity of events below 5% CF





Severity of events below 5% CF, annual and seasonal variation





Worst low wind events

